Linear Diophantine equations

We are interested in finding solutions of systems of equations:

$$\begin{aligned} f_1(x_1, \ldots, x_n) &= 0 \\ \vdots \quad & \quad \vdots \\ f_m(x_1, \ldots, x_n) &= 0 \end{aligned}$$

where $f_I(x_1, \ldots, x_n)$ is a polynomial with *integer* coefficients.

Moreover, we are *primarily* interested in the question of when such equations have *integer* solutions.

Such questions were studied by Diophantus among many others in ancient times and have been give the name *Diophantine problems".

The simplest case is when the polynomials are *linear*. So we are asking when a system of simultaneous linear equations as follows has a solution.

$$\begin{aligned} a_{1,1}x_1 + \cdots + a_{1,n}x_n &= b_1 \\ \vdots \qquad\qquad \vdots \quad & \quad \vdots \\ a_{m,1}x_1 + \cdots + a_{1,n}x_n &= b_m \end{aligned}$$

Here the coefficients $a_{i,j}$ and the values b_i are all integers.

In modern language this can be cast as the problem of determining the image of a map $A : \mathbb{Z}^n \to \mathbb{Z}^m$ given by multiplication by the matrix

$$A = \begin{pmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{pmatrix}$$

Lattices

We can view $\mathbb{Z}^m$ as a subgroup of $\mathbb{R}^m$. Moreover, the latter carries a natural (Euclidean) topology under which $\mathbb{Z}^m$ is a discrete subspace.

Since the image of A is a subgroup of $\mathbb{Z}$, it is also a discrete subgroup of $\mathbb{R}^n$.

A discrete subgroup of a topological group is called a *lattice*.

We can recast our problem as that of finding lattices in $\mathbb{R}^m$.

Note that a subgroup L of $\mathbb{R}^n$ is discrete if and only if there is an $r > 0$ so that the ball $B_r(0)$ of radius r around the origin 0 in $\mathbb{R}^n$ meets L *only* in $\{0\}$. Equivalently, if $v \in L$ such that $\|v\| \leq r$, then $v = 0$.

We claim that there is a linearly independent set $v_1, \ldots, v_k$ of vectors in $\mathbb{R}^n$ so that L is the collection of linear combinations $c_1v_1 + \cdots + c_kv_k$, where c_i are *integers*.

We prove this by induction on m.

If $L = \{0\}$, then the result is trivially true. Note this implies the result when $m = 0$!

When $L \neq \{0\}$, there is a non-zero element $v \in L$ and put $R = \|v\|$. The closed ball $B_R(0)$ is compact. Hence, its intersection $F = L - \{0\} \cap B_R(0)$ with the discrete set $L - \{0\}$ is *finite* and non-empty. If v_1 is an element of F of minimal length, then we see that $0 < \|v_1\| \leq \|v\|$ for *all* v in $L - \{0\}$.

We claim that:

- $L \cap \mathbb{R}v_1 = \mathbb{Z}v_1$;
- $L/\mathbb{Z}v_1$ is a discrete subgroup in $\mathbb{R}/\mathbb{R}v_1$.

This will allow us to complete the proof by induction.

To give a topology on $\mathbb{R}/\mathbb{R}v_1$, we define

$$\|v\|_1 = \inf\{\|v - \alpha v_1\| : \alpha \in \mathbb{R}\}$$

and note that $\|v\|_1 = 0$ if and only if $v \in \mathbb{R}$.

Exercise: Given v, show that there is an integer n and $\alpha \in [-1/2, 1/2]$ such that

$$\|v\|_1 = \|v - (n + \alpha)v_1\|$$

(Hint: Note that $B_R(v)$ is compact for all R so that $F_R = \mathbb{Z}v \cap B_R(v)$ is finite.. For R sufficiently large choose n so that nv_1 is the element of F_R that is closest to v.)

From the triangle inequality, we get

$$\|v - nv_1\| \leq \|v\|_1 + \|v_1\|/2$$

It follows that if $\|v\|_1 < \|v_1\|/2$, then v *cannot* lie in L *unless* $v = nv_1$. In other words,

$$L \cap \{v : \|v\|_1 < \|v_1\|/2\} = \mathbb{Z}v_1$$

Exercise: Use this identity to prove both of the above claims.

Further details can be found on the [relevant page of Wikipedia](https://en.wikipedia.org/wiki/Lattice_(group).

Integer Matrix Reductions

The above argument suggests a way to obtain a more algebraic and algorithmic approach as well.

Given a homomorphism $A : \mathbb{Z}^n \to \mathbb{Z}^m$, the image is unchanged if we make a change of basis of $\mathbb{Z}^n$ as a free abelian group. Moreover, a change of basis of $\mathbb{Z}^m$ *could* lead to a simplified expression of the image.

We will work with:

Column Reduction: In this ccase we pick $i, j \in [1, n]$ and a constant c to make the change of basis:

$$(e_1, \ldots, e_n) \mapsto (e_1, \ldots, e_{j-1}, e_j - ce_i, e_{j+1}, \ldots, e_n)$$

In terms of this basis, the matrix A is replaced by $A(\mathbf{1}_n + E_{i,j}(c))$ where $\mathbf{1}_n$ is the $n \times n$ identity matrix and $E_{i,j}(c)$ is the $n \times n$ matrix with all antries 0 *except* the (i, j)-th entry which is c.

Row Reduction: In this ccase we pick $i, j \in [1, m]$ and a constant c to make the change of basis:

$$(e_1, \ldots, e_m) \mapsto (e_1, \ldots, e_{j-1}, e_j - ce_i, e_{j+1}, \ldots, e_n)$$

In terms of this basis, the matrix A is replaced by $(\mathbf{1}_m + E_{i,j}(c))A$ where $\mathbf{1}_n$ is the $m \times m$ identity matrix and $E_{i,j}(c)$ is the $m \times m$ matrix with all antries 0 *except* the (i, j)-th entry which is c.

Suppose $a_{r,c}$ is the smallest non-zero entry (in absolute value) in the matrix A. If $a_{r,j}$ is any other entry in the same row, we write $a_{r,j} = q_j a_{r,c} + a'_{r,j}$ with $0 \leq a'_{r,j} < |a_{r,c}|$ using integer division. We then apply column reduction to replace the j-th column of A by replacing $a_{i,j}$ by $a_{i,j} - q_j a_{i,c}$. This corresponds to subtracting q_j types the c-th column of A from the j-th column of A. We repeat this for each column $j \neq c$. The entries in the r-th row are non-negative and less than $a_{r,c}$. If one of them is non-zero them we take it to be the *new* $a_{r,c}$.

Repeating this process we ensure that the smallest non-zero entry $a_{r,c}$ in the matrix A is the *only* non-zero entry in the r-th row of A.

We do the same with rows and columns simultaneously to ensure that the smallest non-zero entry $a_{r,c}$ (in absolute value) in the matrix A has the property that all *other* entries in the same row or column are 0.

Suppose $a_{i,j} \neq 0$ for some other i and j. As above, we use division to write $a_{i,j} = qa_{r,c} + a'_{i,j}$. We now perform the following operations:

1. Add q-times the c-th column to the j-th column. Since $a_{i,c} = 0$, we note that $a_{i,j}$ does not change.
2. Subtract the r-th row from the i-th row.

The (i, j)-th entry of the new matrix is $a'_{i,j}$. If it is non-zero, then we have produced a changed matrix with a *smaller* entry than before.

Repeating all the above process we can ensure that the smallest non-zero entry $a_{r,c}$ (in absolution value) *divides* all the entries of the matrix A. Moreover, in the r-th row or c-th column other than $a_{r,c}$ are 0.

We now repeat the process with the matrix obtained by deleting this row and column.

As a result, we see that we an produce a matrix A such that the only non-zero entries are $a_{r_1,c_1}, \ldots, a_{r_k,c_k}$ where the r_i's and c_i's are all distinct. Moreover, a_{r_i,c_i} divides a_{r_j,c_j} for $i < j$.

It is clear how to calculate the image of such a matrix A. By reversing the steps, we can precisely describe the image of the original matrix A. This can be used to give a complete solution of the linear Diophantine problem.

We see as a consequence that if $d_i = |a_{r_i,c_i}$, then the *cokernel* of A is isomorphic to the product of the groups $\mathbb{Z}/\langle d_i \rangle$. This is essentially the fundamental theorem about finitely generated Abelian groups.

Further details can be found on the relevant page of Wikipedia.